QUESTION 1

In embedded systems, why is power management particularly important?

Answer is Correct **Score: 1 / 1**

Options

A. To prevent electromagnetic interference

Correct

B. Because they often run on limited energy sources like batteries or solar cells

C. To avoid writing complex software

D. To reduce manufacturing costs

FEEDBACK

Solution: Many embedded systems rely on limited energy sources that cannot provide continuous or large amounts of power, making power management crucial. Link: https://youtu.be/zce5UqrbGzI?feature=shared&t=16

QUESTION 2

What is the relationship between power, energy, and time when voltage and current are constant?

Not Attempted **Score: 0 / 1**

Options

A. Power = Voltage × Time, Energy = Power / Time

Expected

B. Power = Voltage × Current, Energy = Power × Time

C. Power = Current × Time, Energy = Voltage × Time

D. Power = Voltage / Current, Energy = Power × Current

FEEDBACK

Solution: Power is given by P = V × I, and energy consumed over a time T is given by Energy = Power × Time.

Link: https://youtu.be/zce5UqrbGzI?feature=shared&t=154

QUESTION 3

Which of the following are typical power modes available in many microcontrollers?

Not Attempted **Score: 0 / 1**

Options

Expected
A. Sleep mode

Expected
B. Standby mode

Expected
C. Shutdown mode

D. Turbo mode

FEEDBACK

Solution: Common low-power modes in microcontrollers include sleep, standby, and shutdown, each consuming different levels of power.

Link: https://youtu.be/zce5UqrbGzI?feature=shared&t=431

QUESTION 4

Why does power consumption directly affect thermal design in embedded systems?

Not Attempted **Score: 0 / 1**

Options

A. Higher power consumption increases voltage stability

Expected

B. Higher power consumption means more current draw, which generates more heat

C. Thermal design prevents energy loss through resistance

D. Heat improves the efficiency of embedded processors

FEEDBACK

Solution: The current drawn by the system produces heat, so higher power consumption requires better thermal management to prevent overheating. Link: https://youtu.be/zce5UqrbGzI?feature=shared&t=252

QUESTION 5

What is one strategy mentioned for extending battery life in an embedded system?

Not Attempted **Score: 0 / 1**

Options

A. Run at maximum power continuously

Expected

B. Use short bursts of high power followed by deep sleep modes

C. Avoid switching between power modes

D. Disable all peripherals permanently

FEEDBACK

Solution: A system can run at high power briefly for essential tasks, then enter deep sleep to save energy and extend battery life.

Link: https://youtu.be/zce5UqrbGzI?feature=shared&t=384